ELEC 221: Lab Report Guidelines

- Each individual who is doing the lab **must bring a pre-lab report** to the lab. The pre-lab reports will be **initialized by the TA**.

- Each group must submit a **complete Lab Report (one PDF file per group)** to the OnQ Dropbox.

Section	When to Complete	Notes
Lab report	During and after the lab	Record all experimental steps, results, and observations as a group while performing in the lab. After completing the lab, work together to finish analysis, discussion, and conclusions. **Only one shared lab report is required**, which should include the Experiment section with results.

Note: The **final completed version** of the Lab Report must be submitted to the OnQ Dropbox **by the deadline announced on OnQ**.

Recommendation: We strongly encourage **all group members** to read the lab manual **before coming to the lab**. Having an overview of the procedure in advance will help your group follow the steps more smoothly and understand the purpose behind each part of the experiment.

ELEC221: Electric Circuits
Lab 2 Pre-Lab: Resistive Circuits and Voltage Division

ELEC 221- Fall 2026 **Lab 2 Report**

Group Number:	
Date Performed:	
Section Number:	

Group Members Name	**Group Members Student Number**

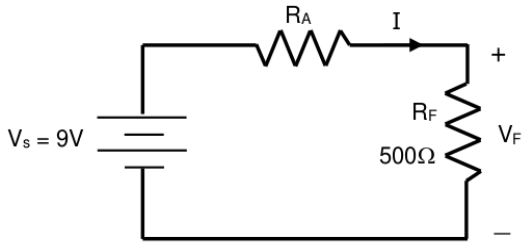
Objectives:

1. Practice analyzing resistive circuits to determine voltage, current, and power.
2. Understand voltage divider behavior and the effect of loading.
3. Apply Kirchhoff's laws to verify current and voltage relationships in circuits.

Definitions:

Resistive Network: A circuit made up of resistors connected in series, parallel, or a combination, used to control current and voltage distribution**.**
Voltage Divider: A circuit with two or more resistors in series used to produce a fraction of the input voltage.
Loading Effect: The change in output voltage of a divider when another resistor is connected in parallel with part of it.
Kirchhoff's Current Law (KCL): At any node, the sum of currents entering equals the sum of currents leaving.
Parallel Resistors: Resistors connected across the same two nodes; their equivalent resistance is smaller than each individual resistor.
Series Resistors: Resistors connected end-to-end; their equivalent resistance is the sum of all resistances.

1) The circuit below is to be used to power a small DC fan motor using a $9V$ battery. R_F represents the fan motor, which requires a minimum current of $10\ mA$ to operate. The power delivered to the motor must be limited to $100\ mW$.
   a) For the values of R_A given in the Table below, determine the current in the circuit I , the voltage across the motor, V_F and the power delivered to the motor P_F.

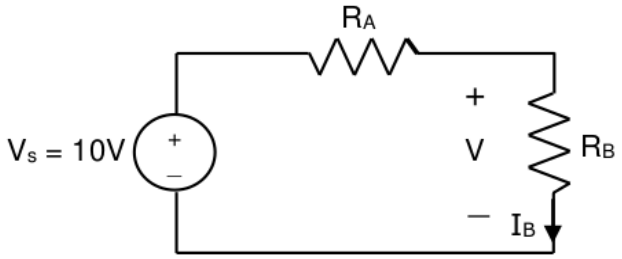


R_A	I	V_F	P_F
0			
100 Ω			
500 Ω			
1 KΩ			

   b) Which values of $R_A$could you use in the circuit? Explain your choices.

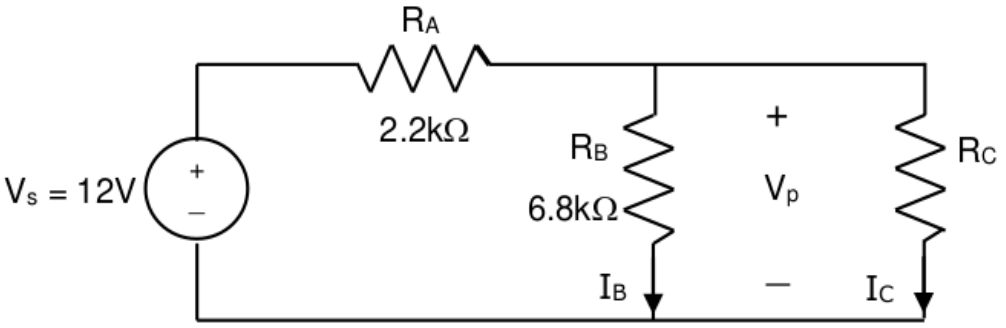
c) What effect does raising R A have on the power delivered to the motor? Explain why this is happening.

2) For the following voltage divider circuit:



a) Find the voltage V_B if $R_A = 2.2\ K\Omega$ and $R_B = 6.8\ K\Omega$.

b) Now, you want to add a resistor, R_C, in parallel with R_B so that the voltage across the parallel combination, $V_P = 5V$. Find the required value of R_C.

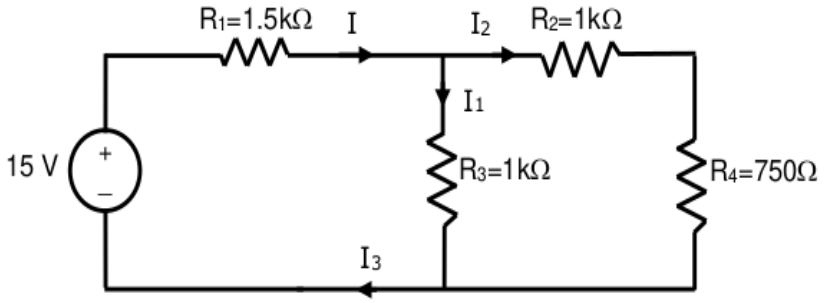


$R_C =$ _____

c) What are the currents through the two branches, I_B and I_C ?

d) What is the power delivered to R_C ?

3) For the circuit shown below:



a) Find the currents, I , I_1, I_2 and I_3.

b) Show that KCL is satisfied for this circuit.

ELEC221: Electric Circuits
Lab 2 (LTspice Experiments): LEDs and the Bridge Circuit

Objectives:

4. How a potentiometer works as a variable resistance
5. How voltage is distributed across series resistors—the voltage divider
6. How to drive an LED
7. How to simulate a circuit in LTSPICE
8. How a bridge circuit works

Definitions:

DMM (Digital Multimeter): A tool used to measure voltage, current, and resistance in a circuit.
Protoboard (Breadboard): A board with holes used to quickly build circuits without soldering.
Potentiometer: A variable resistor with a knob that changes resistance when turned. It is commonly used to adjust things like volume or brightness in a circuit. *For more information see The Audio Pages: Beginners' Guide to Potentiometers. Rod Elliot (ESP). (http://sound.westhost.com/pots.htm)*
Wiper: The movable contact inside a potentiometer that slides along the resistor to change the resistance**.**
Voltage Divider: A circuit with two or more resistors in series used to produce a fraction of the input voltage.
Light Emitting Diode (LED): It is a special type of diode that gives off light when current flows through it in the forward direction. LEDs are part of the diode family, but unlike a normal diode (which only conducts), an LED also emits visible light. They are widely used in indicators, displays, and lighting. You will study LEDs in more detail later in your Electronics course, where you will learn about their characteristics and applications.
LTspice (Linear Technology Simulation Program with Integrated Circuit Emphasis): It is one of the most essential tools for electrical engineers, both in education and industry. In LTspice, electronic components such as resistors, capacitors, diodes, LEDs, transistors, and power sources are modeled with their real electrical properties, enabling users to analyze and test circuits on the computer before building them in hardware. **Bias Point Analysis:** A LTspice analysis that calculates the DC voltages and currents in a circuit when it has settled and is no longer changing with time (this condition is called steady state). **Node Voltage:** The electrical potential at a point in the circuit relative to the ground
Bridge Circuit (Wheatstone Bridge): A bridge circuit is a diamond-shaped network of four resistors powered by a voltage source. It compares voltages at two points to detect balance or imbalance. It is commonly used to measure unknown resistances or sense changes from devices like strain gauges. A condition in a bridge circuit where the resistor ratios are equal and the output voltage is nearly zero is called **Balanced Bridge**.

Part A: Potentiometers and Variable Resistance

The circuit symbol for a potentiometer, often called a pot, is shown in Figure 1.

Fixed resistance, e.g., 10kΩ, between D and E

As the wiper is moved along the length of the resistor, resistance varies from some small value to a maximum of the fixed resistance value.

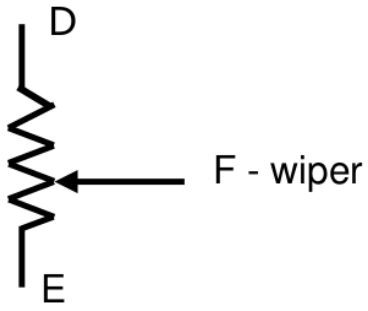


Figure 1: Circuit symbol for the potentiometer.

A potentiometer provides a variable amount of resistance as the wiper is moved along the length of the resistor by turning a knob that is connected to a shaft. Pots are often used to control voltage, by changing the amount of resistance seen by the circuit. A common application is volume control in audio equipment.

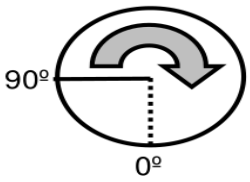
There are two potentiometers at the bottom of the protoboard—a 10kΩ potentiometer on the left and a 1kΩ potentiometer on the right. These are single turn potentiometers—the knob can be turned through about 300º from the minimum to maximum position.

Above each potentiometer is a small breadboard connector. The four outermost holes are connected to the ends of the potentiometer (points D and E); the four inner holes are connected to the wiper (point F). Using the DMM, measure resistance between outer terminals of the 10kΩ potentiometer and record this value here:

Measured Resistance (10kΩ potentiometer, outer terminal):

_____

Considering straight down as 0º (as shown below) set the dial to approximately 90º intervals between the position at which the dial is rotated fully counterclockwise and the position at which it is rotated fully clockwise.



For each position of the knob, measure the resistance between one left outer terminal and one inner terminal; record the resistance values in **Table 1.** Measure the resistance between the right outer terminal and the inner terminal at same dial positions and note these values in the table below.

Knob angle	**Resistance between Left / Inner Terminal**	**Resistance between Right / Inner Terminal**
fully counter clockwise		
90º		
180º		
270º		
full clockwise		

Connect the 10kΩ potentiometer in the circuit shown in Figure 2 using the 5V power supply and a 1kΩ resistor.

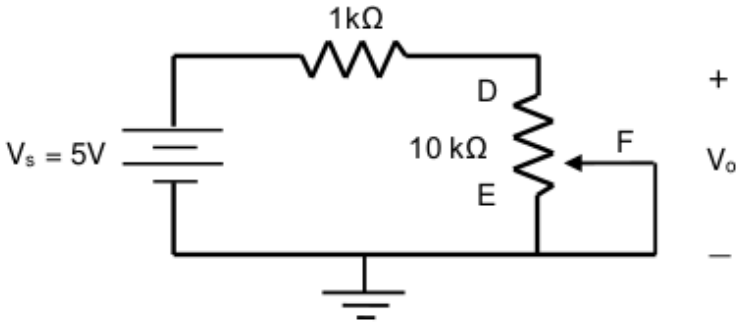


Figure 2: Circuit with variable resistance

Starting with the potentiometer knob rotating fully counterclockwise, use the DMM to measure the output voltage V o for the approximate knob positions given;

Source voltage: $V_S =$			
Knob angle	V_0	Knob angle	V_0
fully counter clockwise			
90º			
135º			
180º			

Powering an LED

LED's or light emitting diodes are semiconductor devices that emit light when electric current is applied. Connect the LED as shown below. The 10K potentiometer will act as a variable resistor; the symbol for a variable resistor is shown on the right in the figure below.

Note: Turn off the power to the protoboard when connecting your circuit. The short leg of the LED is connected to the cathode and should be grounded.

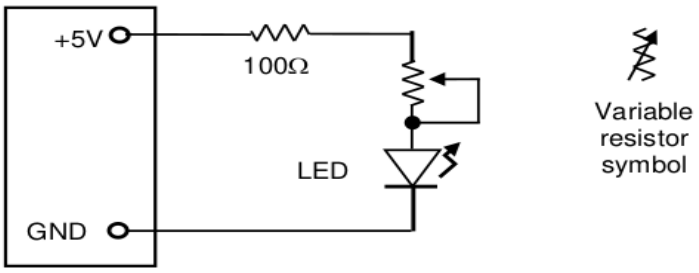


Figure 3: Circuit with a variable resistor connected to an LED.

Circuit with a variable resistor connected to an LED

Vary the potentiometer knob between the point where the LED is just off to where it is fully bright and record the approximate knob angles in the table below, along with the corresponding voltage across the 100□ resistor and the voltage across the LED using the DMM, as shown below. Note the state of the LED—very bright, bright, slightly dimmed, dim, very dim, off. Calculate the current in the circuit from the 100□ resistor voltage drops.

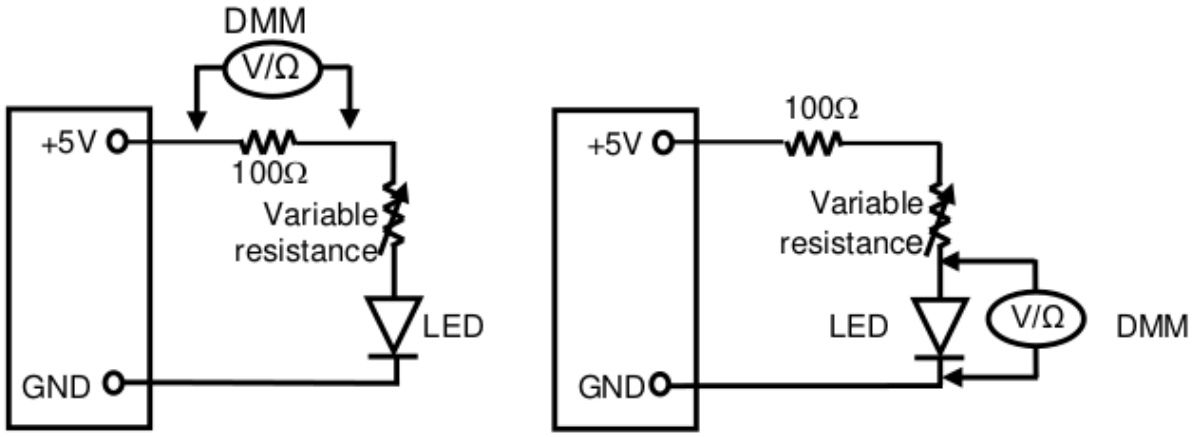
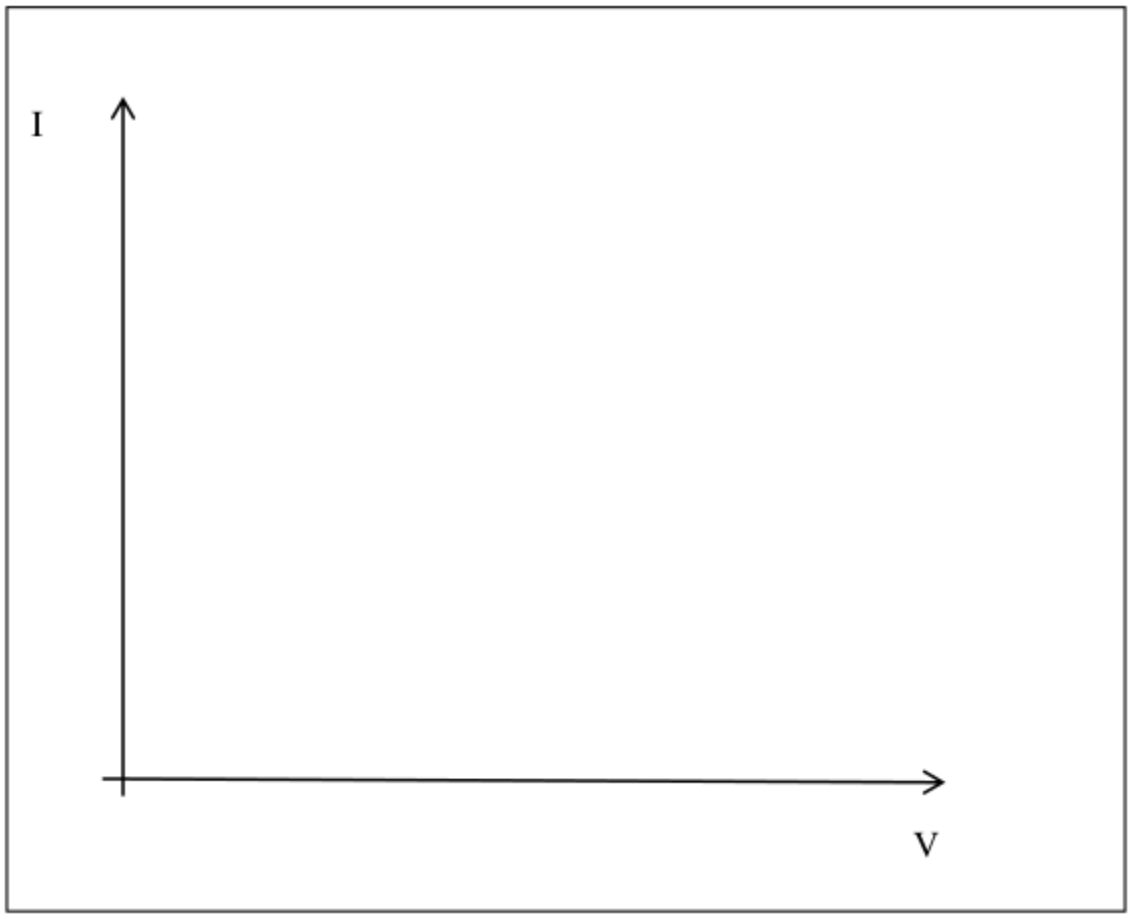


Figure 4: LED circuit measurements

Approximate knob angle	$V_{100}\ (V)$	$V_{LED}\ (V)$	$I(mA)$	$LED\ state$

Sketch the $V - I$ curve of the LED below, labeling the axes.



Part B: Bridge Circuit (LTspice Circuit Simulation)

Attention: In this part, you will design the circuits using LTspice (see Appendix). The TA will begin with an introduction and demonstration. Afterward, you will work in groups to complete Part B. If you have questions, please ask your TA.

You may use the lab computers; if LTspice is not installed, installation instructions are provided in the Appendix. As LTspice is free software, you may also install it on your personal laptop.

Simulate the circuit shown below in LTspice:

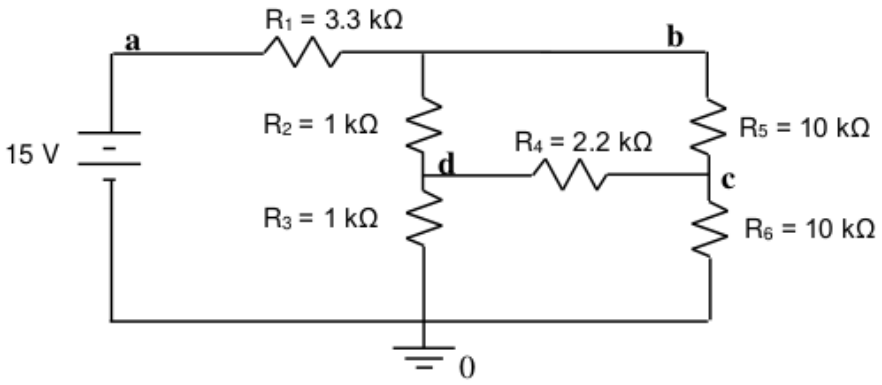


Figure 5: Bridge circuit.

Perform a Bias Point analysis to get the node voltages and currents I_1 to I_6 (*currents through **R1** to **R6** respectively*). Record the voltages at nodes a, b, c and d and the currents, I_1 to I_6 in the table below.

V_a		I_1	
V_b		I_2	
V_c		I_3	
V_d		I_4	
		I_5	
		I_6	

Show your completed LTspice analysis to a TA or the instructor and have him/her note that you completed the exercise.

Validation with Protoboard Experiment: Construct the bridge circuit shown in **Figure 5** on the protoboard. Use the decade resistance for R_6 and set the initial value to 10 kΩ. Use the □5-15V power supply and set the value to approximately 15V using the DMM. Record the measured power supply voltage on the answer sheet. Use the DMM to measure the voltage drop across each resistor. Connect the meter leads so that the red lead corresponds to the first letter in the voltage subscript, e.g., for V_{ab} connect the red lead to **point and** the black lead to **point b**. If the measured voltage is negative, include a negative sign in your recorded value.

Use the DMM to measure the current in each branch of the circuit and record the currents in the table below.

$\boldsymbol{V_{ab}}$		$\boldsymbol{I_1}$	
$\boldsymbol{V_{bc}}$		$\boldsymbol{I_2}$	
$\boldsymbol{V_{bd}}$		$\boldsymbol{I_3}$	
$\boldsymbol{V_{cd}}$		$\boldsymbol{I_4}$	
$\boldsymbol{V_c}$		$\boldsymbol{I_5}$	
$\boldsymbol{V_d}$		$\boldsymbol{I_6}$	

Build Fig.6 circuit and change the value of $R_6$to the values given in the table below and fill in the corresponding values of the voltage across the middle resistor.

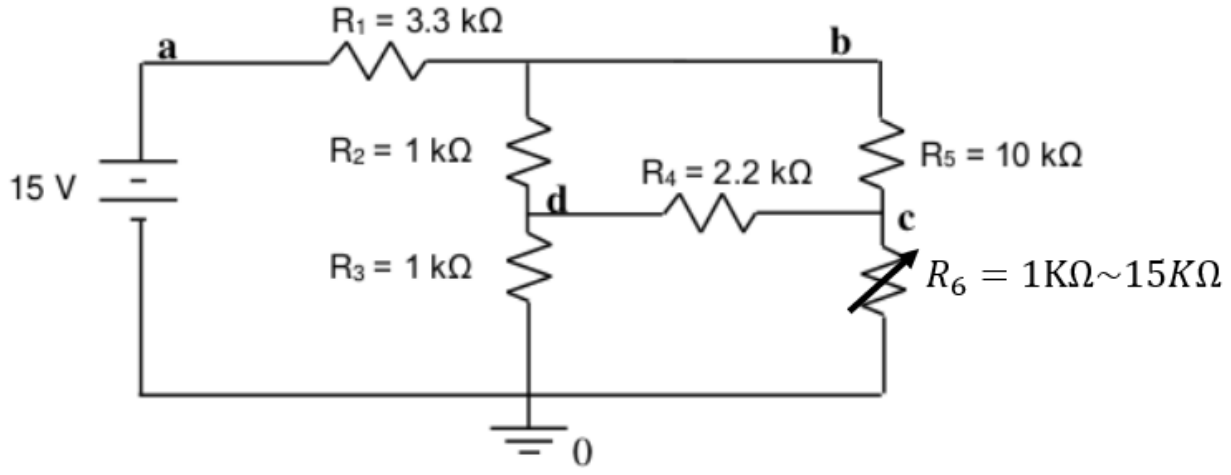


Figure 6: Bridge circuit with potentiometer.

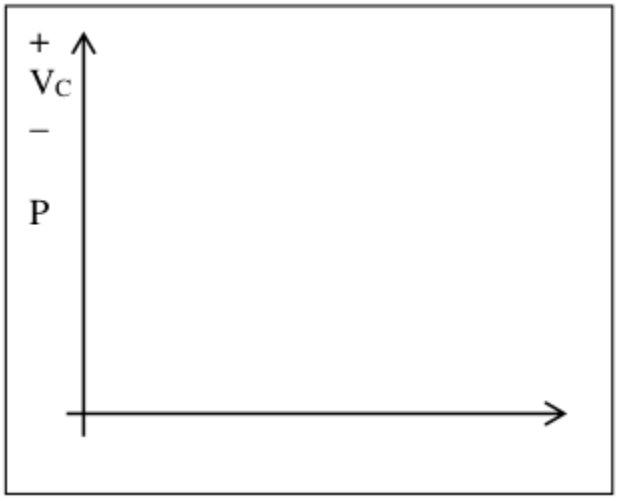
R_6 (Ω)	V_{cd} (V)	R_6 (Ω)	V_{cd} (V)
5K		**11K**	
6K		**12K**	
7K		**13K**	
8K		**14K**	
9K		**15K**	
10K			

Now that you have completed the Experiment section, please work with your group to answer the following questions.

These follow up questions are designed to help you analyze and reflect on your experimental results.

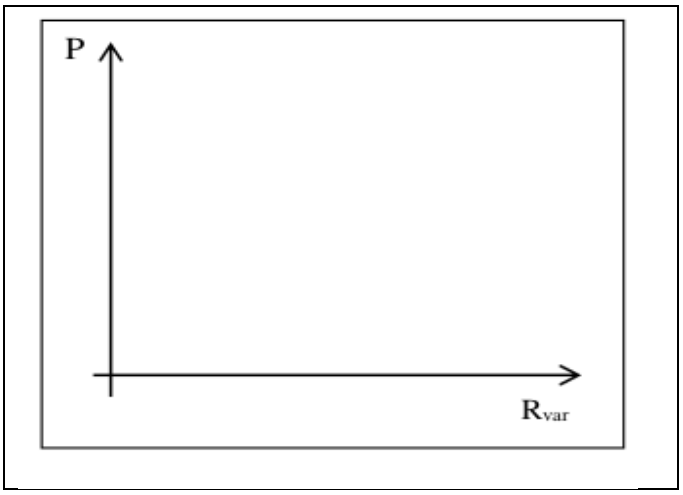
1. **Voltage divider:** For the voltage divider circuit, calculate the current in the circuit and the power delivered to the variable resistance, for each knob angle. Plot power versus the variable resistance.

Knob Angle	V_o	I	P
Full CCW			
90º			
180º			
270º			
Full CW			



+ V_R -

2. **LED Network:** Plot the power delivered to the LED versus R var below.



Does the LED act like a resistor?

Why?

3. **Bridge Network:** How do the simulated voltages and currents compare to the measured values?

4. If the **2.2kΩ** resistor were replaced by a current meter, how might the resulting circuit be used?

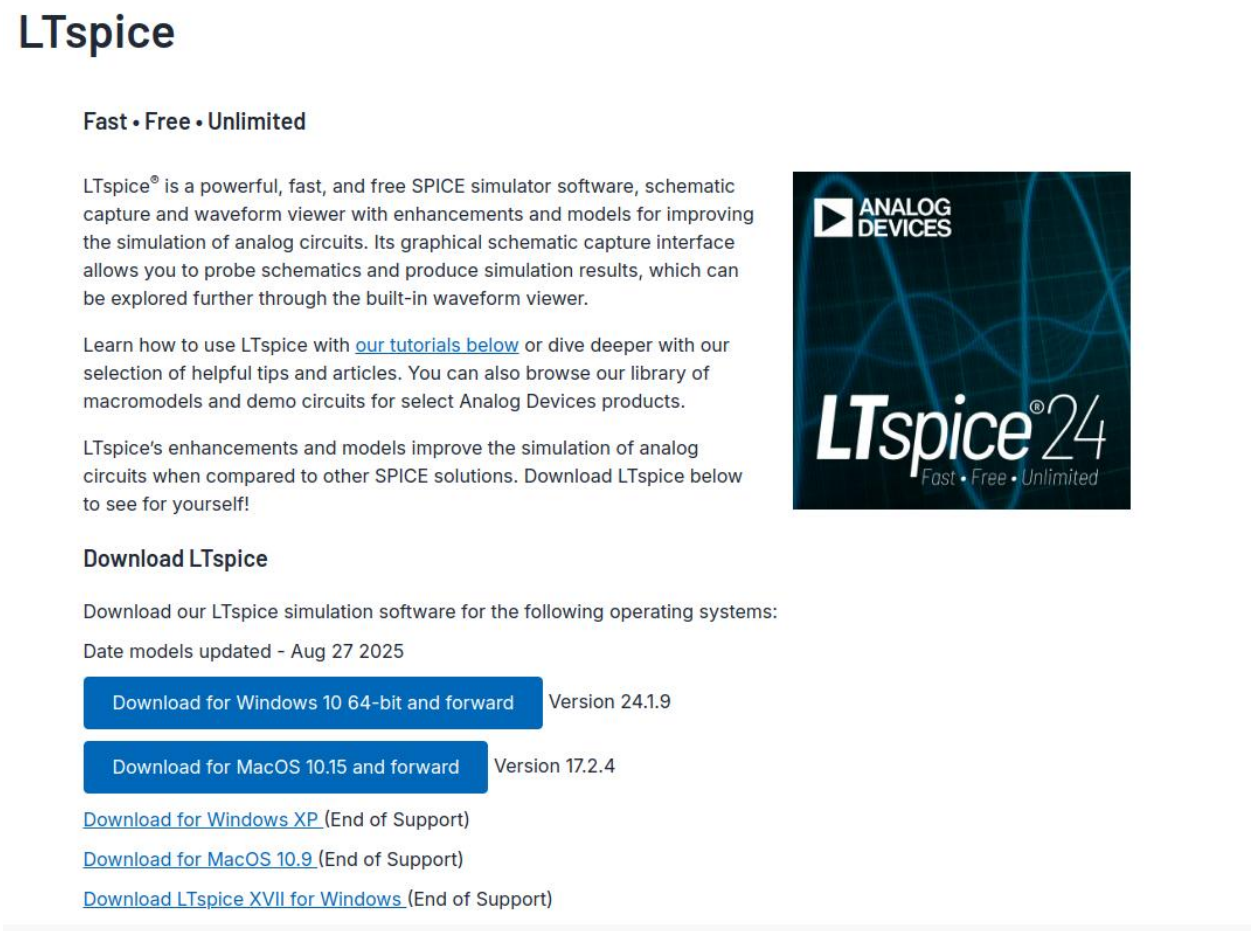
5. **Strain Gauges.** What are the sources of error and how much might each contribute to any discrepancy?
   Consider in particular the supply voltage and strain gauge "**at rest**" resistances.

ELEC221: Electric Circuits
Appendix 1- Windows only
LTspice Basics

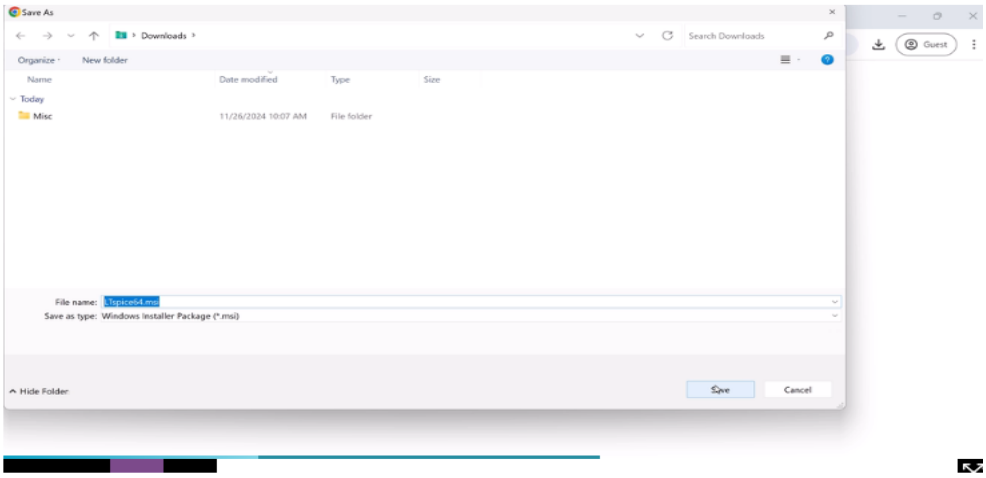
Downloading and Installing:

To download LTspice, go to: analog.com/ltspice

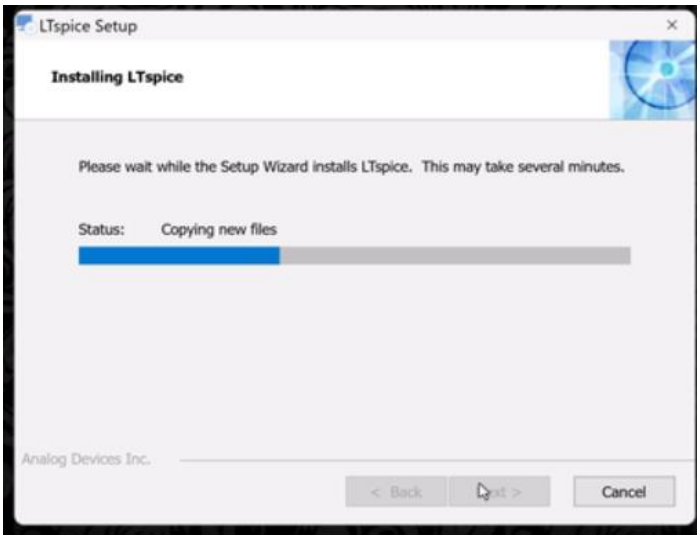
1. **Choose the correct configuration for your desktop computer:**



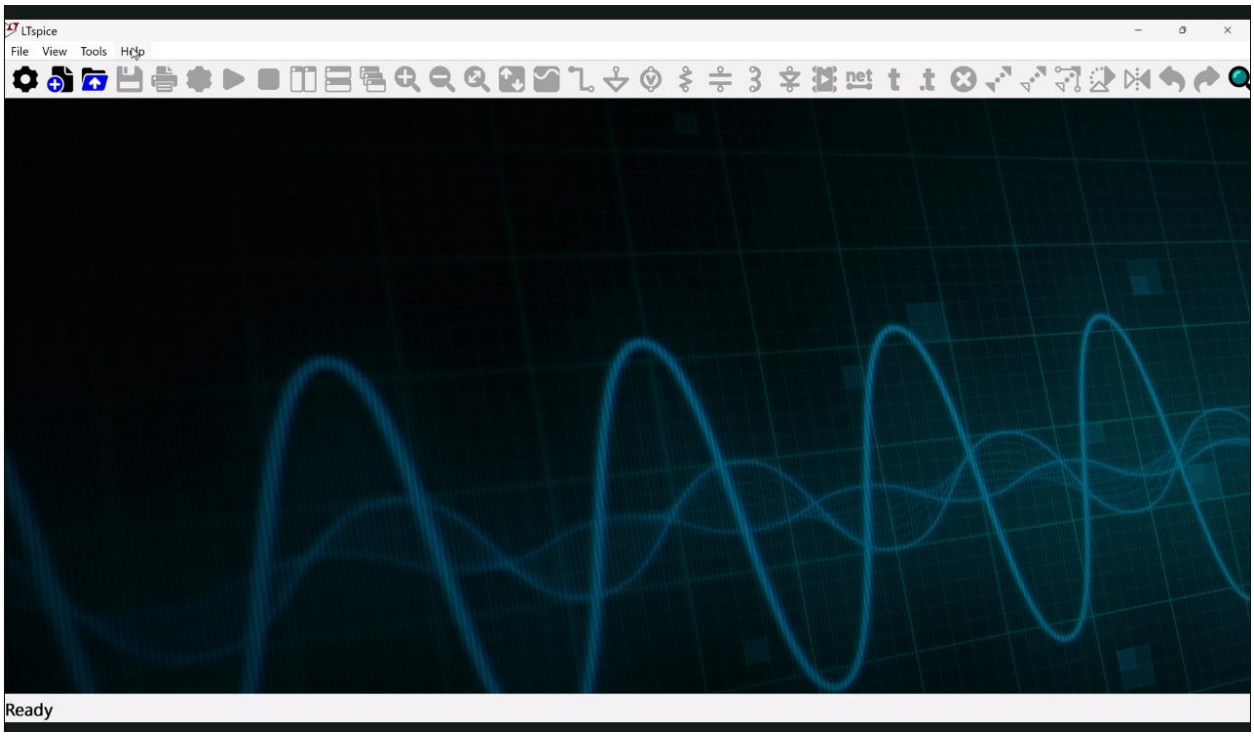
2. **Download the LTspice:**



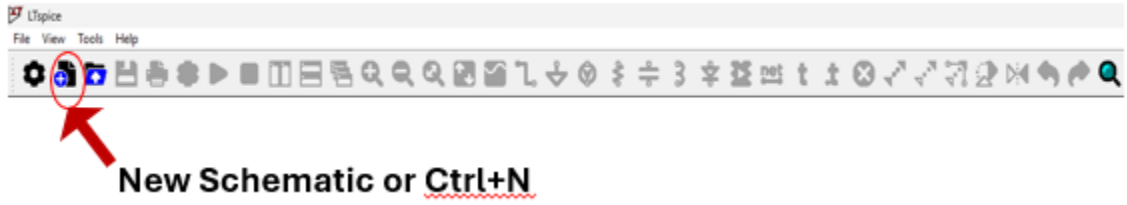
3. **Install the LTspice:**



4. **Lunch the LTspice:**



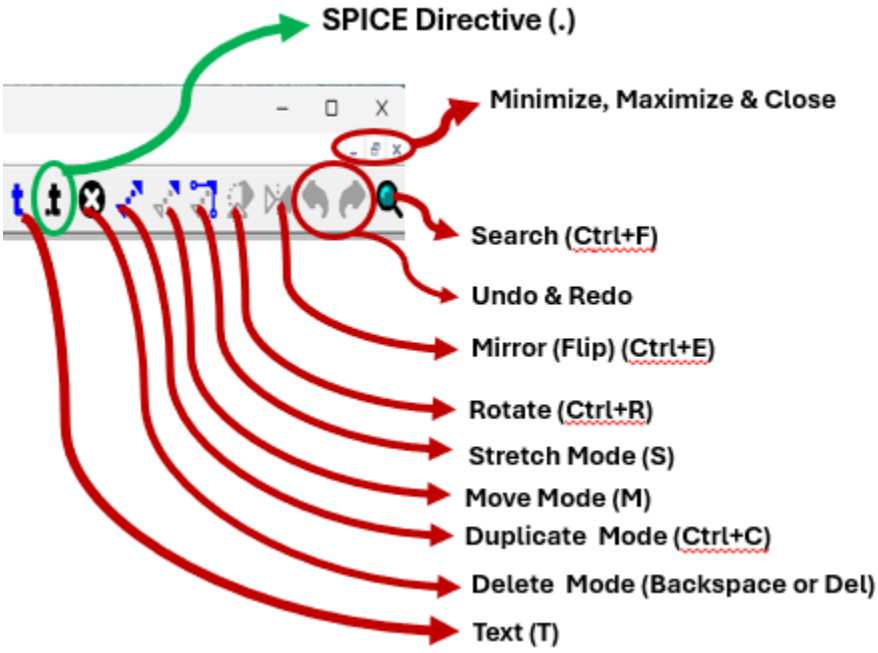
5. **Click the toolbar icon (circled) below to create New Schematic (saved as Draft.asc) and view the activated elements**



See the activated elements after creating your new schematic:

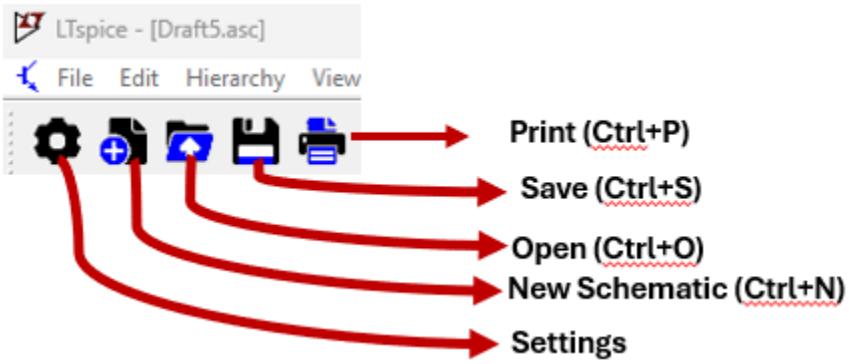


6. **Description of the right-side toolbar icons:**

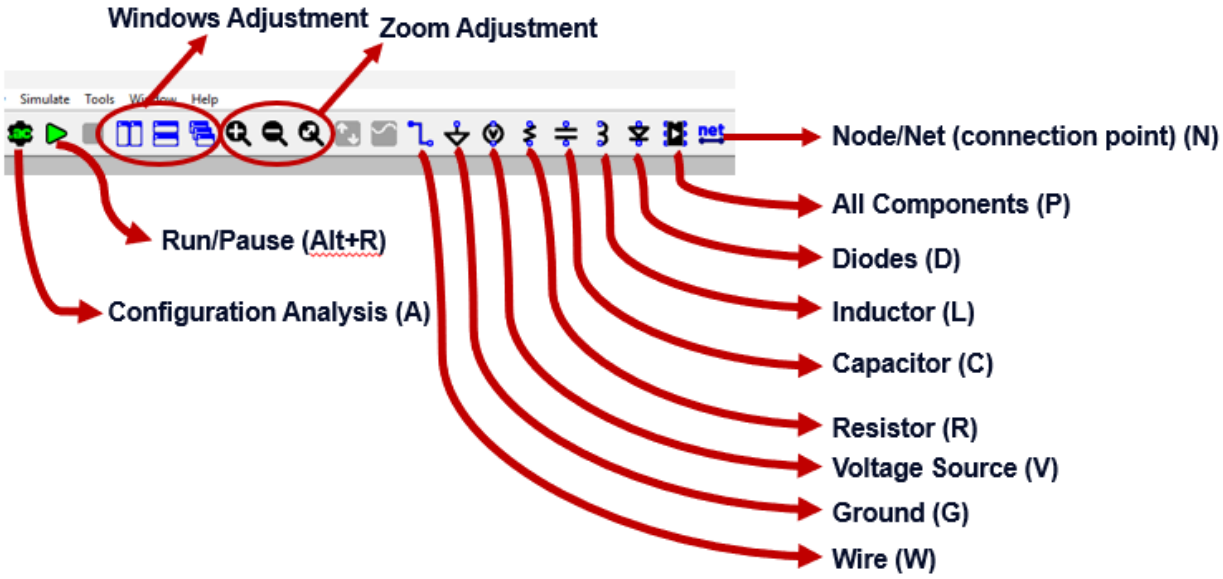


SPICE Directive (.): A text command in LTspice that tells the simulator how to run, such as defining parameters, sweeping values, or setting initial conditions.

7. **Description of the left-side toolbar icons**



8. **Description of the middle (main) toolbar icons:**



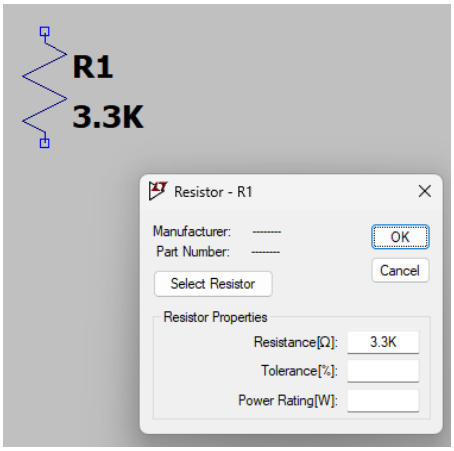
Node/net (connection point) (N): on the schematic can be labeled by using the Label Net tool button or F4 key.

Mouse Actions in LTspice

Action	On Component	On Wire	On Empty Space / Waveform
Right Click (single)	Open component value/properties window **stop from many active actions**	Show wire label / properties	Select tool / choose trace in waveform viewer

In LTspice, when you select a component (for example, a resistor using **the "R" hotkey**) and place it with **a left click**, the component symbol remains active so you can continue placing additional copies. Once you have placed as many components as you need, you **simply right click** or press **the Escape key** to stop placement, so the tool does not keep adding more of the same component.

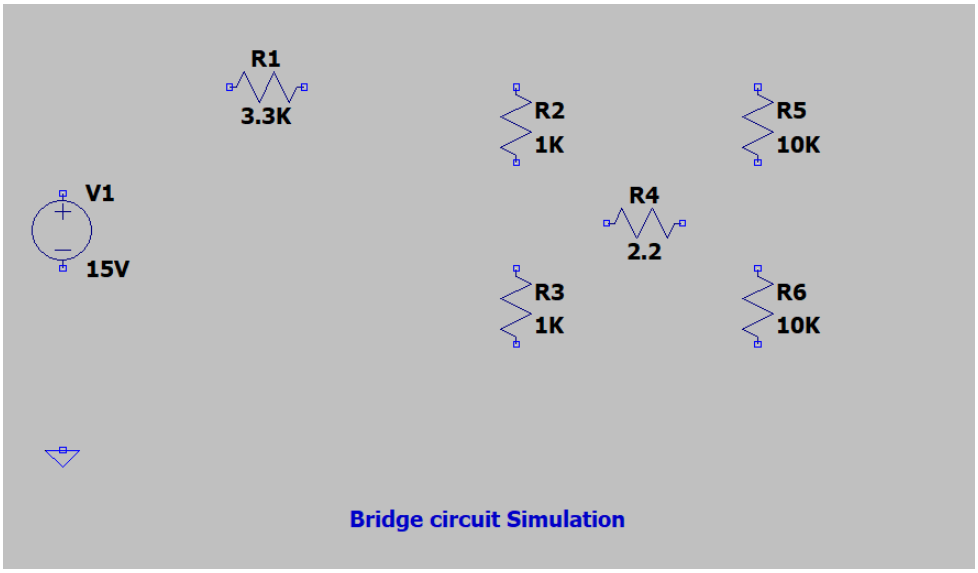
Hint: You can set the value of any component by typing it directly into its properties window.



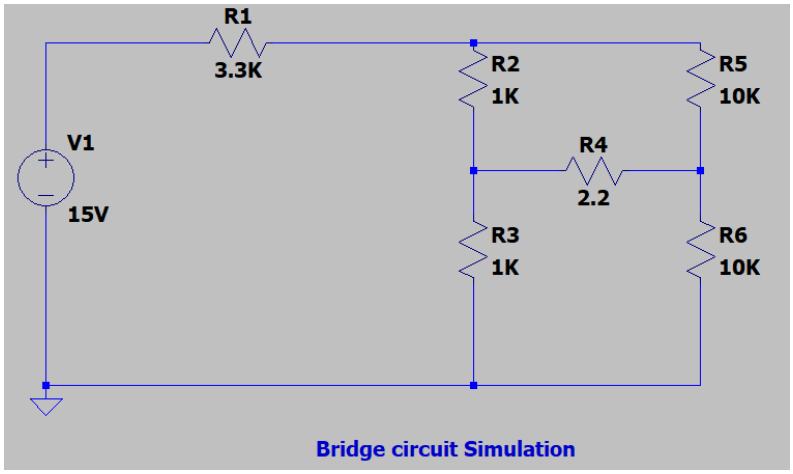
Moving, Rotating and Mirroring Components in LTspice

Action	Steps	Shortcut
Activate Move Mode	Click the **Move** tool on the toolbar or press **M**	**M**
Move a Component	**In Move mode**, left click the component to select it, then **drag it to the new position**	**---**
Rotate a Component	**While in Move mode**, select the component, then press **Ctrl + R** to rotate clockwise	**Ctrl + R**
Mirror (Flip) a Component	**While in Move mode**, select the component, then press **Ctrl + E** to mirror it	**Ctrl + E**
Stop Moving/Rotating	Press **Esc** or **right click** to exit placement/move	**Esc**

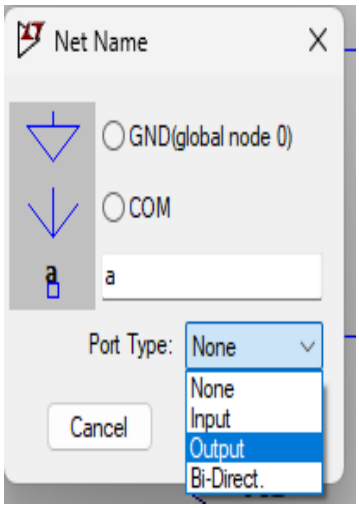
9. **Your group Bridge Circuit design should follow these steps:**
   a) **Place all components (respectively,R1,R2,...), assign their values and rotate them if necessary.**
   Attention 1: In LTspice, every circuit needs a **ground** (you should place it) because the simulator requires a reference point **(0 V**) to calculate all node voltages.

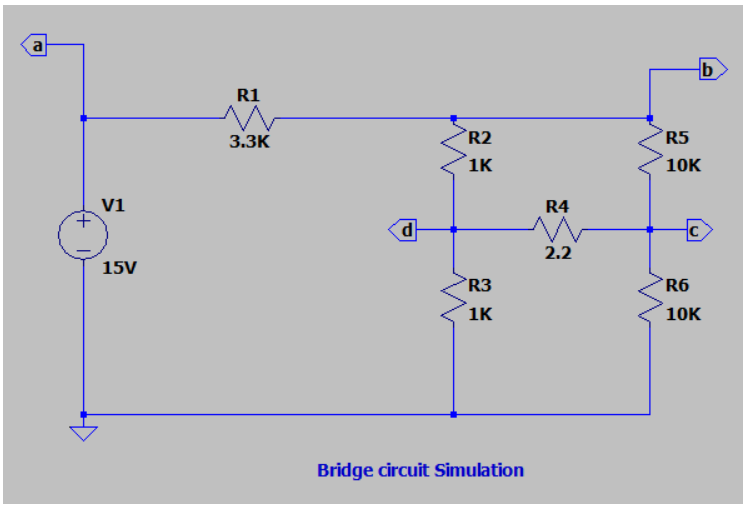


   b) **Wire them:**



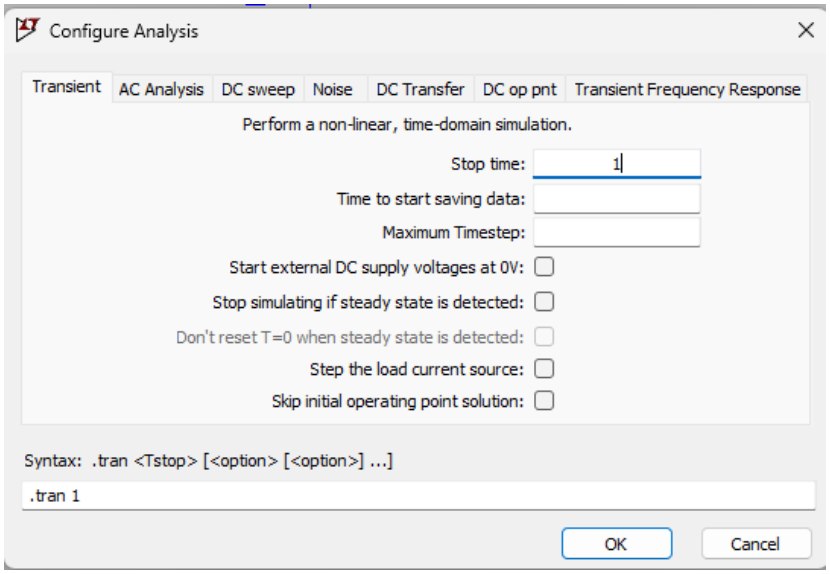
c) Place nodes a, b, c and d using the Net Label tool:



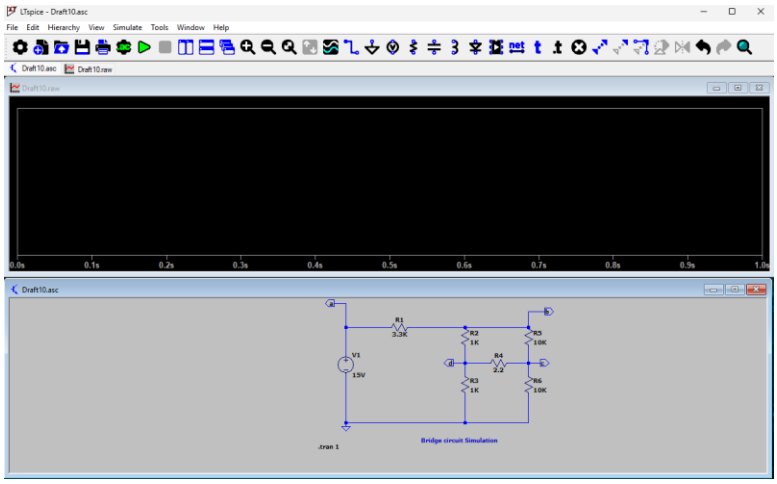


d) Click on the RUN toolbar icon and configure Analysis:

set the **stop time** to 1 (second) so that the directive **(.tran 1**) appears in your group design.



e) Then your analysis result will be saved as (.raw) file:



f) Placing the **red probe** on any node will display the voltage at that point. If you hover the mouse cursor over a component, LTspice will display **the current flowing through it (Blue Prob)**.

g) To calculate the voltage difference between two nodes (e.g., Vab), define it mathematically in the analysis configuration.

Hint: You can delete or remove the displayed result at any time.

